Recall-Constrained Alert Triage Across SOCs: Statistical Threshold Calibration

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***Abstract*—Security operations centers (SOCs) receive far more alerts than analysts can review, and triage tools are usually judged by a single score, F1, on one dataset. We examine whether that is sufficient, using alerts from a Wazuh deployment that we attacked ourselves (969 labelled alerts) and the public AIT Alert Data Set (eight simulated enterprise scenarios). Three findings emerge. First, a high F1 can be obtained by flagging nearly every alert, which saves analysts no work; at realistic attack rates such systems are close to flagging everything. Second, results do not carry over between environments: the ranking of an LLM, a rule-level threshold and trained classifiers differed between the datasets, and thresholds chosen in one environment often missed the recall target in another. Third, a classical order-statistic tolerance bound can select a threshold from a small random sample of analyst-labelled alerts, with at least 90% recall at 90% confidence under stated assumptions. In simulations on public alert streams totalling about 390,000 alerts, it reached an average coverage of 96.5% in the settings where it could certify, at the cost of clearing fewer benign alerts than a naive threshold. A live implementation shows that the calibration workflow is feasible and that the threshold is sensitive to labelling errors. The results support evaluating at realistic attack rates, recalibrating for each environment, and checking labels before relying on a recall guarantee.**

***Index Terms*—alert triage, security operations center, SIEM, alert fatigue, threshold calibration, recall guarantee, LLMs**

I. Introduction

When detection rules are triggered, SIEM systems raise alerts. The volume of alerts is too high for the number of analysts, and most alerts are false positives, which leads to alert fatigue and missed attacks [1]. Mitigations either adjust the SIEM's own risk scoring and thresholds or use machine learning [2]–[5] and, more recently, LLMs [6] for the triage decision. These methods are typically tested with F1 or accuracy on one dataset [7] and then deployed in an environment with a different attack rate, hosts and logs. We ask: (i) does a high F1 on a test set with many attacks actually mean that the analysts' workload is reduced? (ii) Does the ranking of the methods (rule-level threshold, trained classifiers, an LLM) hold across datasets? (iii) When a model moves to a new environment, how should its threshold be set, and what can be said about the attacks it catches? We also test whether LLMs, which are costly to run, beat cheap baselines.

Contributions. (1) An analysis, with projection to realistic attack rates, showing that a high F1 can come from flagging everything. (2) A replication on the public AIT data [8], [9], where an LLM is on par with trained models and far above the rule-level threshold, unlike on our data. (3) Evidence on both datasets that thresholds do not transfer. (4) An evaluation of a classical distribution-free threshold rule (a tolerance bound [10], as in the Neyman–Pearson umbrella algorithm [11]) for alert triage: coverage on 391,488 alerts against a naive alternative, labelling cost, and a negative result for active label selection. The rule itself is not new. (5) A live deployment of the calibration workflow.

II. Related Work

Pietraszek and Tanner [12] studied learning-based alert post-processing, and Feng et al. [2], Ban et al. [3], Kapera and Niemiec [4] and Uetz et al. [5] study ML-based triage, volume reduction and risk-based alerting on proprietary, production or multi-dataset alert data (Table I). Nonparametric tolerance limits date to Wilks [10]; the Neyman–Pearson umbrella algorithm [11], risk-controlling prediction sets [13] apply such distribution-free ideas to classifiers, but not to alert triage. To our knowledge, none of the triage studies projects results to realistic attack prevalence, measures how fixed thresholds behave across environments, or measures how often a few-label rule meets its recall target and at what cost.

III. Data and Setup

Our deployment and data. Wazuh 4.9.2 on a Google Cloud VM monitors a second VM running a deliberately vulnerable web app (SQL and OS-command injection, XSS, template injection). Rounds 1–3 used hand-crafted attacks (manual SSH guessing, curl-based web attacks); round 4 used real tools (hydra, gobuster, nikto, sqlmap), which trigger Wazuh correlation rules the manual rounds never did. The dataset has 969 labelled alerts in 26 categories, with labels assigned by us from rules and known attack scripts (no independent annotator). We train on rounds 1–3 (273 alerts, 6 source IPs) and test on round 4 (696 alerts, one unseen IP, 82.3% attacks); the split is time-, IP- and methodology-disjoint. **Features and models.** Each alert is described by its Wazuh severity level (rule_level), the hour it occurred, its rule group and decoder, and the words in its log line (TF-IDF, up to 300 terms, single words and word pairs). IP addresses, ports and timestamps in

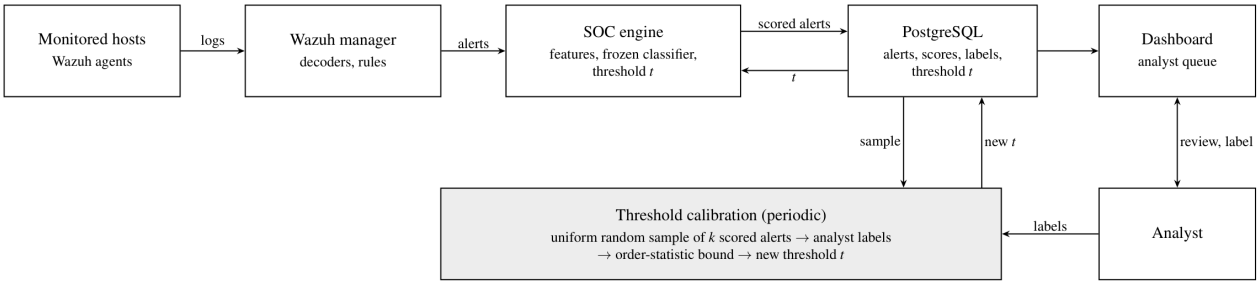


Fig. 1. System architecture. Wazuh alerts are scored by a frozen classifier and gated by the threshold t; PostgreSQL stores alerts, scores, labels and the threshold. Periodically, a uniform random sample of scored alerts is labelled by an analyst through the dashboard and the order-statistic bound writes a new t.

TABLE I
WHAT EACH RELATED STUDY EVALUATES, COMPARED WITH THIS PAPER (FROM THE CITED PAPERS' ABSTRACTS AND REPORTED RESULTS).

Work	Data	Main evidence reported	Shift or guarantee
Feng et al. [2]	one production SOC	ML workflow from labels to risk scores	not addressed
Ban et al. [3]	one real-world network	cost-sensitive learning and event segmentation for IDS alerts	not addressed
Kapera, Niemiec [4]	one production deployment	26% fewer false and repeated alerts	dynamic risk thresholds; no cross-environment test
Uetz et al. [5]	eight alert datasets	mean AUROC 0.92 (combined hypotheses) vs. 0.72 (severity only)	ranking quality only
Tong et al. [11]; Bates et al. [13]	general classifiers, not alerts	high-probability error control (theory)	guarantees, but not for alert triage
This paper	own Wazuh deployment (969 alerts) and AIT-ADS (8 scenarios, 391,488 alerts)	prevalence-projected F1; threshold transfer; coverage and label cost of a recall-guaranteed threshold	transfer measured; classical guarantee tested under its assumptions

the log text are first replaced by placeholders so that models cannot use them as shortcuts. We compare seven methods: Random Forest (RF; 300 trees, depth 12), logistic regression (LogReg), a small neural network (MLP; layers of 64 and 32 units), XGBoost (300 trees, depth 6), a simple cut-off on rule_level (chosen with Youden's J on the training data), an ensemble that combines the RF with a recent-alert count and the severity level, and a large language model (LLM) prompted with and without examples and never shown the true labels. RF, MLP and XGBoost depend on a random seed, so we repeat them (20 runs for RF, 10 for the others) and report mean $\pm$ standard deviation; with only 273 training alerts their results vary widely. Fig. 1 shows the pipeline studied throughout and Fig. 2 how its threshold is calibrated.

Public data. The AIT Alert Data Set [8], [9] has real Wazuh alerts from eight simulated enterprise scenarios (2,600,263 alerts). We label an alert an attack if and only if its timestamp lies in an attack-phase window published with the data set (noisy, window-level labels); no alert-type override is applied, so any alert inside a window, including scanner "400 error" alerts, is an attack here. We use leave-one-scenario-out evaluation: train on seven scenarios, test on the eighth.

Reproducibility. The source code, the 969-alert labelled dataset, the live-calibration labels and corrections, and every experiment script with its saved output are maintained on GitHub (https://github.com/vizarrd/recall-constrained-alert-triage) and archived on Zenodo (doi:10.5281/zenodo.[number]). A reproduction guide maps each table and figure to its script, with fixed random seeds and exact package versions (Python 3.11, scikit-learn 1.9.1). The public AIT data set is not redistributed; it is obtained from its Zenodo record [8] and processed by the provided sampling scripts. The LLM runs used blind sessions of a hosted model and may differ on rerun; raw verdicts are included.

IV. EVALUATION PITFALLS ON OUR DATA

Table II shows round 4 with every method at its default threshold. Because 82.3% of the round-4 alerts are attacks, simply calling every alert an attack already gives F1 0.903. The severity threshold (rule_level), the LLM and LogReg score about the same (0.905–0.909), but only because they also flag almost everything: they clear just 2–7% of the harmless alerts. Real streams contain far fewer attacks [14], so we re-estimate each method's results for a lower attack share π from how often it catches attacks (TPR) and raises false alarms (FPR), assuming both stay the same as the attack share changes, using $P(\pi) = \pi\,\mathrm{TPR}/(\pi\,\mathrm{TPR} + (1-\pi)\,\mathrm{FPR})$ for precision. At $\pi = 5\%$ these methods reach only F1 0.10, and an analyst must check about 19 alerts to find one real attack, almost the same as the 20 needed when everything is flagged (XGBoost is worse: 33). A high F1 therefore hides the problem only when attacks are common.

Ranking versus operating point. Besides the default-threshold results, we judge how well each method separates attacks from noise at any threshold, using the Triage Relief Curve: the share of noise cleared against the share of attacks

TABLE II

ROUND 4 (TRAIN ROUNDS 1–3), DEFAULT THRESHOLD. "CLEARED" = BENIGN ALERTS IDENTIFIED AS BENIGN. F1@5% AND REVIEW BURDEN (ALERTS REVIEWED PER REAL ATTACK) ARE PROJECTED TO 5% ATTACK PREVALENCE, ASSUMING TPR AND FPR STAY UNCHANGED. MLP REVIEW BURDEN IS UNDEFINED (SOME SEEDS FLAG NOTHING).

Method	F1	Recall	Cleared	AUC	Relief@R95	F1@5%	Review
rule_level	.905	1.00	.02	.966	.024	.097	19.5
LLM (few-shot)	.905	1.00	.02	.618	.024	.097	19.5
LogReg	.909	1.00	.07	.969	.935	.101	18.8
XGBoost	.625	.55	.08	.565	.081	.057	33.1
Random Forest	.47±.12	.32±.09	.94±.01	.404±.005	.06±.01	.265	4.7
Ensemble	.67±.01	.51±.01	.95	.556	.07±.00	.420	2.8
MLP	.65±.27	.56±.31	.81±.37	.88±.19	.69±.42	.416	–

still caught. Its summary, Relief@R95, is the noise cleared while keeping at least 95% of attacks. This is just a mirrored ROC curve, not a new method. The RF clears 94% of noise at threshold 0.5, but its ROC-AUC is 0.404, below chance: it calls most unfamiliar traffic benign (recall 0.32) instead of telling the classes apart. The ensemble beats the RF in all 20 seeds (McNemar $p < 10^{-39}$ on seed 0), but that is a better operating point on a weak ranker (AUC 0.556), not better ranking. LogReg ranks well (AUC 0.969; it clears 93.5% of noise while keeping 95% of attacks); the MLP is unstable across seeds. Training on round 4 and testing on rounds 1–3 gives the same picture: LogReg is best (AUC 0.917) and the RF is poor (0.591).

Ensemble. Following Uetz et al. [5], we combined three signals with a logistic regression trained on rounds 1–3: the RF probability, the number of alerts from the same source IP in the previous 10 minutes, and rule_level. Our first attempt failed (F1 0.15, single run) because this count averages 28 in the training rounds but 342 in the test round, a $12\times$ shift; applying $\log(1+x)$ fixed it (F1 0.67 ± 0.01, vs. 0.47 ± 0.12 for the RF).

LLM. We classified all 696 round-4 alerts with an LLM, given the same fields as our classifier (rule level and description, decoder, raw log) and blind to ground truth, in seven independent sessions. A few-shot prompt (told most alerts are benign, eight examples, a probability output) leaves its decisions unchanged (TPR 1.00, FPR 0.976, AUC 0.618).

Label convention. Of the 123 round-4 benign alerts, 107 are one alert type ("Web server 400 error code", scanner noise) that we labelled benign by convention. The RF clears all 107 but only 58% of the other 16; the LLM and rule-level threshold clear none. If scanner noise is labelled *attack* instead, the rule-level threshold and LLM become essentially correct (recall 1.00), the RF's recall drops (0.32 to 0.27) and AUCs change (rule level 0.97 to 0.85, LogReg 0.97 to 0.72), so round-4 method comparisons are convention-dependent.

V. CROSS-ENVIRONMENT BEHAVIOUR ON PUBLIC DATA

Table III shows results on the public data. Each of the eight scenarios is held out in turn and the models are trained on the other seven, using a random sample of 4,000 alerts per scenario (6.8–88.6% attacks). One caveat applies throughout: an alert counts as an attack if it falls inside a published attack time window, not because anyone checked that the alert itself is malicious.

TABLE III

AIT-ADS, MEAN OVER 8 HELD-OUT SCENARIOS. "TRANSFER": THRESHOLD FROM OUT-OF-FOLD TRAINING SCORES FOR 95% RECALL. "+20": RE-THRESHOLDED WITH 20 LABELLED TARGET ALERTS.

Model	AUC	Relief@R95	Transfer recall / cleared	+20 recall / cleared
rule_level	.692	.339	.94 / .69	.94 / .54
LogReg	.960	.695	.84 / 1.00	.86 / .86
Random Forest	.964	.735	.81 / 1.00	.93 / .84
MLP	.957	.660	.92 / 1.00	.87 / .83

LLM versus trained models. Methods that read the log content beat the rule_level threshold clearly (AUC 0.96 versus 0.69). The LLM was run on a smaller sample of 150 alerts per scenario (1,200 alerts in total), taken from the 4,000-per-scenario sample behind Table III, so its scores cannot be compared with that table or with results on all 32,000 alerts. For a fair comparison we therefore scored the trained models (each trained on the other seven scenarios) on exactly the same 1,200 alerts. The mean AUC per scenario is 0.959 for the LLM, against 0.950 for LogReg, 0.954 for RF and 0.947 for MLP; the rule-level threshold is far behind. To check whether these small gaps are real, we resampled the alerts and the scenarios many times (bootstrap). The 95% intervals for the LLM's advantage are narrow and mostly include zero against the trained models (for example -0.009 to $+0.027$ against LogReg), but are large against the rule-level threshold ($+0.22$ to $+0.31$). The LLM beats LogReg, RF and MLP in only 4, 2 and 3 of the 8 scenarios, and beats the rule-level threshold in all 8. We therefore do not claim a winner: the LLM is on par with the trained models and clearly better than the rule-level threshold. It also does not simply flag everything; on our own data it did so because of the labelling convention described above. Removing feature groups from LogReg shows why the datasets differ: on the public data no single group matters much (AUC stays at 0.958–0.962; the severity level alone gives only 0.692), while on our data the severity level alone gives 0.966, dropping it gives 0.397 and the log text alone 0.334 (below chance). The public data is also easier: 90% of its attack-labelled alerts are the "Web server 400 error" alert, which *our own* dataset labels benign, so the two datasets use opposite conventions for scanner noise.

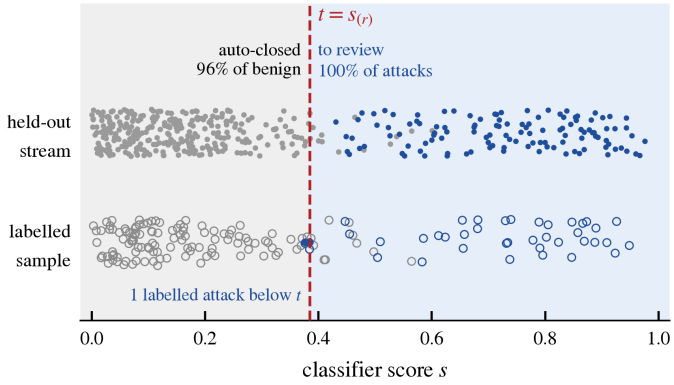


Fig. 2. How the threshold is obtained (illustrative seeded simulation, not experimental data): $k = 160$ uniformly drawn alerts are labelled ($n = 39$ attacks; filled dots mark the r lowest). The r-th lowest labelled attack score is the threshold t; held-out alerts below it are auto-closed.

Thresholds do not transfer. On our data, a 95%-recall threshold chosen on rounds 1–3 makes LogReg and rule_level flag everything on round 4 (noise cleared 6.5% and 2.4%) because attack prevalence moves from 20% to 82% [15]. On AIT, transferred thresholds clear nearly all noise but miss the recall target (0.81–0.92 on average; RF 0.00 on shaw). Re-thresholding with 20 random labels helps the worst case (RF on shaw: 0.00 to 0.82) only partly.

VI. A Recall-Guaranteed Threshold

Rule. Draw k alerts at random and have an analyst label them; n of them turn out to be attacks, with model scores $s_{(1)} \le \cdots \le s_{(n)}$ (sorted). Let ρ be the share of attacks we want to catch (target recall) and δ the chance we allow of missing that target. Choose the largest rank $r \le n$ for which $\Pr[\mathrm{Beta}(r, n - r + 1) \le 1 - \rho] \ge 1 - \delta$, set the threshold to $t = s_{(r)}$ (the r-th lowest attack score), and flag every alert that scores at least t. If no such r exists, there are too few labelled attacks, so we flag everything ($t = -\infty$, the *fallback*). Certification needs at least $\lceil \ln \delta / \ln \rho \rceil$ labelled attacks (22 for $\rho = 0.9$, $\delta = 0.1$), which is about $22/p$ labelled alerts when a share p of the alerts are attacks. This is a classical one-sided tolerance bound [10], as in the Neyman–Pearson umbrella algorithm [11] (there it bounds type-I error, here the miss rate); our contribution is applying and evaluating it for alert triage. *Intuition* (Figs. 2 and 3): the lowest scores among the labelled attacks show where the attack scores start. The more attacks are labelled, the higher a rank r we can pick while still being 90% sure that at most 10% of all attacks score below t.

Guarantee. *Assume (A1) the k labelled alerts are a uniform random sample of the alert stream, (A2) the analyst's labels are correct, and (A3) the model and its scores do not change afterwards. Then, for the number n of labelled attacks we have, the recall at threshold t is at least ρ with probability at least $1 - \delta$.* The reason is standard. If the n labelled attack scores are independent draws from all attack scores, the share of attacks scoring below $s_{(r)}$ follows a $\mathrm{Beta}(r, n - r + 1)$ distribution, and recall at t is at least one minus that share. Tied scores can only raise recall, never lower it. In our experiments the labelled alerts are instead drawn without replacement from a finite stream, where the exact probabilities follow a hypergeometric law. We still use the Beta rule, because in practice the number of attacks in the stream is unknown. To check it, we computed the exact success probability of the rule under this sampling for 8,775 settings (n from its minimum up to 1,000; $\rho \in \{0.80, 0.90, 0.95\}$; $\delta = 0.10$; streams with 3,426, 10,000 or 44,524 attacks), assuming no tied scores; it never fell below 0.9007. This is a numerical check on that grid, not a general proof, and the coverage experiments below remain the direct empirical test. The guarantee concerns the attacks in the stream under (A1)–(A3), not every future deployment: if the data drift, the rule must be re-run (we did not test this). It covers recall only; noise cleared is an empirical outcome.

Experiments. We keep three things apart: the mathematical guarantee (which holds under A1–A3), how often it actually held in simulated labelling runs (this section), and whether it holds in real deployments, which we have not established (Section VII). A *trial* draws k random alerts, labels them, sets the threshold with the rule, and checks whether the true recall on the whole stream is at least ρ. *Coverage* is the share of trials in which this succeeds. A trial with fewer than 22 labelled attacks cannot be certified and falls back to flagging everything, which trivially gives recall 1. We therefore check the guarantee on certified trials only (Table IV) and also report the numbers with all trials included. In a first study (LogReg, 100 trials), a naive rule that keeps the fraction ρ of the labelled attacks reached recall 0.90 in only 0.58–0.83 of trials with 20–100 labels, and choosing the alerts to label more cleverly did not help (stratified 0.60–0.84, uncertainty 0.50–0.75).

Large-scale check and provenance. The public data holds 2,600,263 alerts; from each scenario we drew up to 50,000 uniformly without replacement (391,488 distinct alerts; russellmitchell in full), and used 1,000 random labelling trials for each of 240 settings (8 scenarios, 2 models, 5 values of k, 3 values of ρ). Recall and noise cleared are true values on the whole held-out stream. The alerts are time-correlated; this does not affect the guarantee, whose target is the uniformly sampled finite stream, but the evidence rests on eight scenarios.

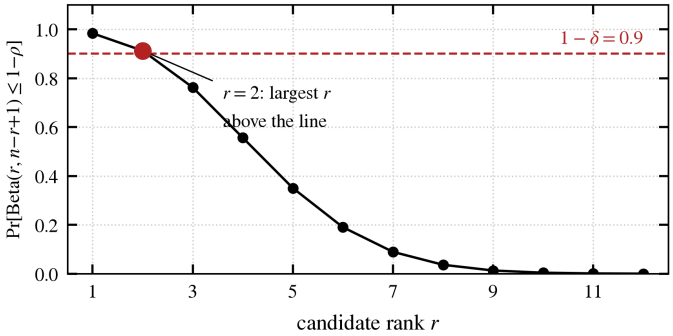


Fig. 3. Choosing r: the largest rank for which $\Pr[\mathrm{Beta}(r, n-r+1) \le 1-\rho] \ge 1 - \delta$ ($n = 39$, $\rho = 0.9$, $\delta = 0.1$).

TABLE IV

COVERAGE AND COST OF THE GUARANTEED THRESHOLD ON THE LARGE AIT-ADS STREAMS (LOGREG; $\rho = 0.90$, $\delta = 0.10$; 1,000 TRIALS PER SCENARIO AND k). $\hat{c}$: SHARE OF CERTIFIED TRIALS (N_{cert}) WITH TRUE RECALL $\geq \rho$, WITH 95% WILSON INTERVAL; $\hat{c}_{\text{all}}$ ALSO COUNTS UNCERTIFIED TRIALS (FLAG ALL) AS SUCCESSES; UNCERT.: SHARE OF TRIALS WITHOUT A CERTIFICATE. GUARANTEE-CHECK COLUMNS USE THE S SCENARIOS WITH ≥ 100 CERTIFIED TRIALS AT THAT k; COST COLUMNS AVERAGE ALL 8 SCENARIOS AND MUST NOT BE COMBINED WITH THEM. †TRIVIAL. BOLD: INTERVAL ABOVE 0.90.

		Guarantee check (same S scenarios)					Cost (all 8 scenarios)	
Rule	k	S	N_{cert}	$\hat{c}$ [95% CI]	$\hat{c}_{\text{all}}$	Uncert.	Benign cleared	Uncert.
Flag everything	–	–	–	1.000†	–	–	0.00	–
Naive threshold	200	8	8,000	0.525 [0.514, 0.536]	0.525	0.00	0.87	0.00
Naive threshold	1000	8	8,000	0.619 [0.608, 0.630]	0.619	0.00	0.87	0.00
Guaranteed	50	4	4,000	**0.961** [0.955, 0.967]	0.961	0.00	0.46	0.50
Guaranteed	100	5	4,425	**0.940** [0.933, 0.947]	0.947	0.12	0.52	0.45
Guaranteed	200	6	5,182	**0.944** [0.937, 0.950]	0.952	0.14	0.56	0.34
Guaranteed	500	8	7,989	**0.952** [0.947, 0.956]	0.952	0.00	0.66	0.00
Guaranteed	1000	8	8,000	**0.955** [0.950, 0.959]	0.955	0.00	0.72	0.00

Results. Fig. 4 shows the picture per scenario. Counting uncertified trials as successes changes the coverage by at most 0.008 (Table IV; largest raw difference 0.0076, at $k = 200$), so the certified numbers are not an artifact of leaving trials out. We look at the 184 of 240 settings that have at least 100 certified trials. The target is 0.90, and the average certified coverage is well above it: 0.965 for the 90% recall target (62 settings), 0.983 for the 80% target (72 settings) and 0.967 for the 95% target (50 settings; lowest 0.906). Two settings fell slightly below 0.90: 0.899 at the 90% target (RF on shaw, $k = 200$; 95% Wilson interval [0.84, 0.94], 148 certified trials) and 0.862 at the 80% target (LogReg on santos, $k = 100$; [0.79, 0.91], 116 trials). Neither interval lies entirely below 0.90; we keep both results. Because they looked low, we reran them with 10,000 trials, a fresh training sample, and recall measured on alerts that were not in the labelled sample. The reruns gave 0.936 [0.92, 0.95] (1,565 certified trials) and 0.930 [0.92, 0.94] (1,363). We reran these two settings precisely because they looked low, which favours higher rerun values, so the reruns support, but do not prove, that the dips were random sampling noise. These are simulation estimates, not a proof.

Is the guarantee worth its cost? Compare three ways of achieving the same goal (recall 0.90). Flagging everything always meets it but clears nothing. The naive rule clears 0.87 of the benign alerts (analysts still review 13%), but it meets the target in only about half of the trials (coverage 0.32–0.80), as expected for a rule that aims exactly at the target. The guaranteed rule achieves a certified coverage above 0.9, but clears only 0.46–0.72 of the benign alerts (analysts review 28–54%). The guarantee is therefore worth paying for when missing an attack is costly and enough labels are available. It needs many labels when attacks are rare: at least $22/p$ labelled alerts at attack share p (2,200 at 1%, 440 at 5%, 220 at 10%); in our 7–9% scenarios about 500 were needed. The naive rule is cheaper if occasional misses are acceptable.

VII. LIVE DEPLOYMENT

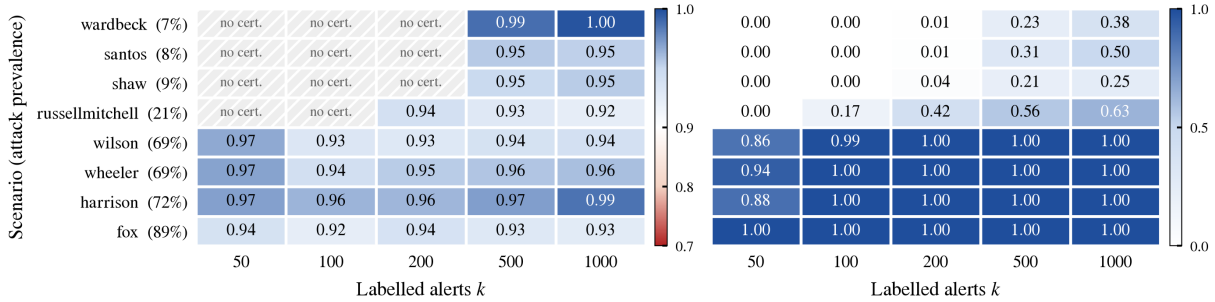
We deployed the system on Google Cloud (Wazuh, the classifier, PostgreSQL and a web dashboard). One analyst labelled randomly selected alerts in the dashboard over two rounds, covering attacks we launched and real internet SSH bots. Until 22 attacks have been labelled, the system reports itself as *uncalibrated* (default threshold 0.5, no guarantee); thereafter it applies the recall-guaranteed threshold of Section VI.

Round 1. With a LogReg model trained on our 969 alerts, 43 labels (36 attacks, after we corrected three routine package-update alerts that had been marked as attacks) certified a threshold of 0.0004, so nearly every alert was flagged. The lowest-scoring labelled attack was a real SSH-bot event ("PAM: User login failed") that the model scored 0.0004, because our training data labelled such events benign. The guarantee held, but exposed a blind spot in the model. Retraining with the 43 live labels (weighted 5) raised the lowest held-out attack score from 0.0004 to 0.022 in cross-validation, but held-out ROC-AUC fell (0.795 to 0.767): the blind spot was removed without improving ranking.

Round 2. With the retrained model and fresh labels, the analyst marked all 28 sampled alerts as attacks, including seven routine "PAM login session opened/closed" events that we classify as benign. We corrected six of these (originals kept) and left the seventh. The system certified at 22 attacks with a threshold of 0.0665, set by that remaining routine alert, and flagged 211 of 229 scored alerts (92%), against 156 (68%) at the default threshold of 0.5. Because the weakest labelled attack sets the threshold, one routine alert labelled as an attack removes almost all noise reduction, and the guarantee needs correct labels (A2). Nine of 71 live labels (13%) required correction. The deployment shows that the loop runs end to end with only a few dozen labels, and that the lowest-scoring labelled attacks act as a diagnostic: they point directly to what the model misses (Round 1) and to labels worth re-checking (Round 2).

VIII. DISCUSSION AND LIMITATIONS

Our comparison does not show that LLMs are better or worse than trained classifiers: their ranking differed between datasets. Likewise, the coverage results come from simulation and should not be read as evidence of production reliability,



(a) Coverage of recall ≥ 0.9 among certified trials (blue $\geq$ 0.9 target, red below)

(b) Benign alerts cleared at the guaranteed threshold

Fig. 4. The guarantee study per scenario (LogReg, $\rho = 0.90$, 1,000 labelling trials per cell; scenarios sorted by attack prevalence). (a) Coverage among certified trials: all certified cells shown are ≥ 0.90 (blue); hatched: too few labelled attacks ($n < 22$) to certify. (b) Benign alerts cleared at the guaranteed threshold: near zero for rare-attack scenarios, about 1 for attack-heavy ones.

nor of recall under distribution shift, label noise or changed attacker behaviour.

Limitations. Both datasets are Wazuh-based and use rule-based or window-level labels. Our own dataset has seven source IPs, a narrow benign class (107 of 123 alerts are of one type) and labels from a single party, while the public dataset is comparatively easy: rule type nearly determines the label. The results should therefore not be assumed to generalize to other SOC environments. The labels were not independently validated. The guarantee assumes random samples, correct labels and no drift; in deployment we tested it with a single analyst only. These limits define the scope of our claims.

IX. Conclusion

We asked whether the usual ways of judging alert-triage tools still hold when a tool moves to a new environment. On our own Wazuh deployment and on the public AIT data, they often do not. First, a high F1 can simply mean that almost every alert is flagged; at realistic attack rates such a system gives analysts nearly as much work as flagging everything. Second, results do not travel: the ranking of an LLM, a rule-level threshold and trained classifiers differed between the two datasets, and a threshold chosen in one environment missed attacks in another. Third, a classical order-statistic rule can set the threshold from a small random sample of analyst-labelled alerts. Once enough attacks are labelled (22 for a 90% target), its recall guarantee held in 96.5% of simulated labelling trials on average over streams of 391,488 alerts, at the price of clearing fewer harmless alerts and of needing hundreds of labels when attacks are rare. The guarantee relies on random samples, correct labels and a stable model; in our live deployment 13% of labels needed correction, so we show the workflow, not a validated live guarantee. In practice we therefore suggest reporting results at realistic attack rates, re-calibrating the threshold with a few random labels whenever the environment changes, and checking labels before trusting the guarantee. Testing the approach on larger, independently labelled production alert streams is the main next step.

References

[1] B. A. Alahmadi, L. Axon, and I. Martinovic, "99% false positives: A qualitative study of SOC analysts' perspectives on security alarms," in *31st USENIX Security Symp.*, 2022, pp. 2783–2800.

[2] C. Feng, S. Wu, and N. Liu, "A user-centric machine learning framework for cyber security operations center," in *Proc. IEEE ISI*, 2017, pp. 173–175, doi:10.1109/ISI.2017.8004902.

[3] T. Ban, T. Takahashi, S. Ndichu, and D. Inoue, "Breaking alert fatigue: AI-assisted SIEM framework for effective incident response," *Appl. Sci.*, vol. 13, no. 11, p. 6610, 2023, doi:10.3390/app13116610.

[4] A. Kapera and M. Niemiec, "Dynamic risk thresholds for SIEM alerting based on machine learning," *IEEE Access*, vol. 13, pp. 121034–121047, 2025, doi:10.1109/ACCESS.2025.3588441.

[5] R. Uetz, P. Bönninghausen, L. Hackländer-Jansen, and M. Henze, "Can risk-based alerting mitigate cybersecurity alert fatigue?" arXiv:2609.02465, 2026.

[6] H. Xu *et al.*, "Large language models for cyber security: A systematic literature review," arXiv:2405.04760, 2024.

[7] D. Arp, E. Quiring, F. Pendlebury, A. Warnecke, F. Pierazzi, C. Wressnegger, L. Cavallaro, and K. Rieck, "Dos and don'ts of machine learning in computer security," in *Proc. 31st USENIX Security Symp.*, Boston, MA, USA, Aug. 2022.

[8] M. Landauer, F. Skopik, and M. Wurzenberger, "AIT Alert Data Set," Zenodo, 2023, doi:10.5281/zenodo.8263181.

[9] M. Landauer, F. Skopik, and M. Wurzenberger, "Introducing a new alert data set for multi-step attack analysis," in *Proc. CSET*, 2024, doi:10.1145/3675741.3675748.

[10] S. S. Wilks, "Determination of sample sizes for setting tolerance limits," *Ann. Math. Statist.*, vol. 12, no. 1, pp. 91–96, 1941.

[11] X. Tong, Y. Feng, and J. J. Li, "Neyman–Pearson classification algorithms and NP receiver operating characteristics," *Sci. Adv.*, vol. 4, no. 2, p. eaao1659, 2018.

[12] T. Pietraszek and A. Tanner, "Data mining and machine learning—towards reducing false positives in intrusion detection," *Inf. Secur. Tech. Rep.*, vol. 10, no. 3, pp. 169–183, 2005, doi:10.1016/j.istr.2005.07.001.

[13] S. Bates, A. Angelopoulos, L. Lei, J. Malik, and M. I. Jordan, "Distribution-free, risk-controlling prediction sets," arXiv:2101.02703, 2021.

[14] S. Axelsson, "The base-rate fallacy and the difficulty of intrusion detection," *ACM Trans. Inf. Syst. Secur.*, vol. 3, no. 3, pp. 186–205, Aug. 2000.

[15] M. Saerens, P. Latinne, and C. Decaestecker, "Adjusting the outputs of a classifier to new a priori probabilities: A simple procedure," *Neural Comput.*, vol. 14, no. 1, pp. 21–41, 2002, doi:10.1162/089976602753284446.